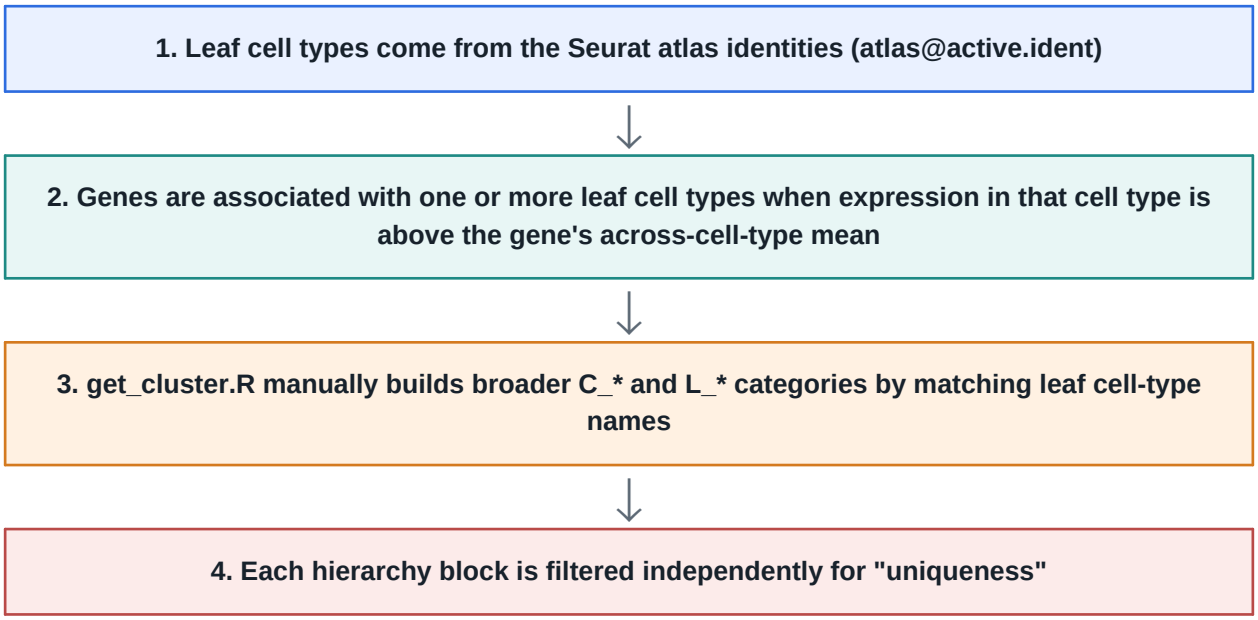
How the Hydra cell-type hierarchy works

From leaf cell types to C_* superclusters and L_* categories, with the exact uniqueness logic used in get_cluster.R and analyze_deseq.R



Core idea

The hierarchy is **not inferred by a clustering algorithm**. The leaf annotations are inherited from the atlas, and the broader categories are explicitly defined in code. A gene can therefore belong to several leaf cell types and several broader categories at the same time before the uniqueness filter is applied.

Level	Examples	How it is created
Leaf cell type	En_Tentacle, Ec_Tentacle, Ec_BasalDisk, Ec_Peduncle	Original atlas cell identities; gene assignment based on expression criterion
C_* superclass	C_tentacle, C_EC_foot, C_neuron	Manual grouping of leaf names in get_cluster.R
L_* category	L_foot, L_ECTODERM, L_ENDODERM, L_INTERSTITIAL	Manual broader grouping in get_cluster.R

Repository basis: Kanomble/hydra_clustering - scripts/dropseq_final_atlas_analysis.R, scripts/create_sc_atlas_cluser_table.ipynb, scripts/get_cluster.R, scripts/analyze_deseq.R.

1. How a supercluster is created

get_cluster.R starts with a gene x leaf-cell-type TRUE/FALSE matrix. Then it defines broader categories with explicit name-matching rules.

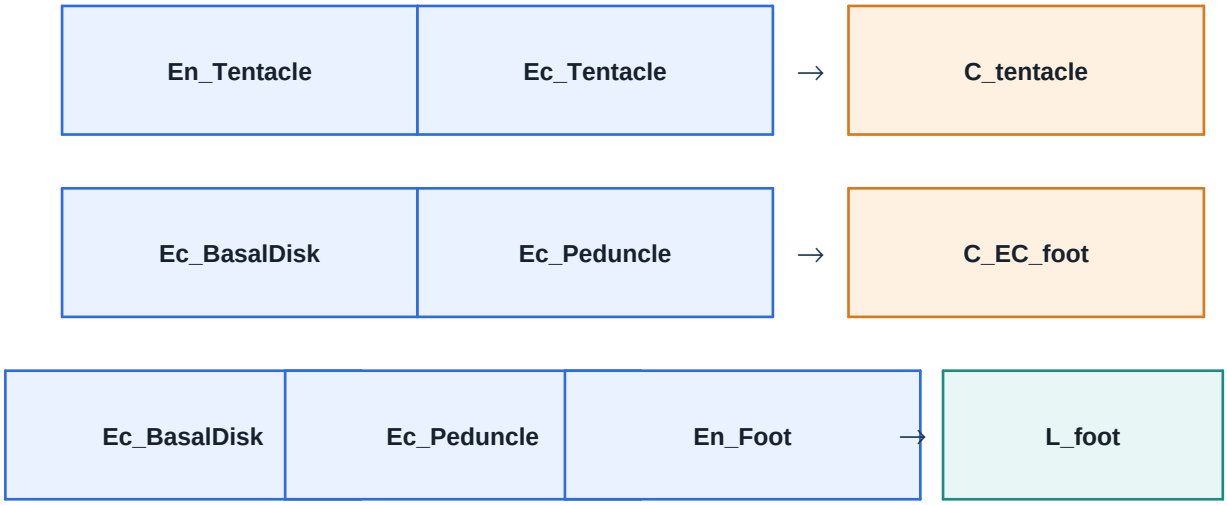
Example definitions from get_cluster.R

```
super[["C_EN_foot"]] <- grep("En_Foot", cids, value=TRUE)
super[["C_tentacle"]] <- grep("_Tentacle$", cids, value=TRUE)
super[["C_EC_foot"]] <- c(grep("Ec_BasalDisk", cids, value=TRUE),
grep("Ec_Peduncle", cids, value=TRUE))
super[["L_foot"]] <- c(grep("Ec_BasalDisk", cids, value=TRUE),
grep("Ec_Peduncle", cids, value=TRUE),
grep("En_Foot", cids, value=TRUE))
```

Membership rule

For every manually defined category, a gene is positive if it is present in **any** child cell type belonging to that category:

```
apply(ctab[, x, drop = FALSE], 1, any)
```



Important: the C_* and L_* annotations do not replace the leaf annotations. They are added as additional columns. Therefore a gene may initially carry both leaf-level and superclass-level assignments.

2. Worked example: one gene in four leaf cell types

Assume Gene X passes the expression criterion in these four atlas cell types:

Because superclass membership uses OR / any(), Gene X also receives:

Derived category	Why Gene X is assigned	Value
C_tentacle	En_Tentacle or Ec_Tentacle is positive	**1**
C_EC_foot	Ec_BasalDisk or Ec_Peduncle is positive	**1**
L_foot	Ec_BasalDisk or Ec_Peduncle is positive	**1**

Before uniqueness filtering, Gene X can therefore look like this:

Leaf level	En_Tentacle	Ec_Tentacle	Ec_BasalDisk	Ec_Peduncle
Value	**1**	**1**	**1**	**1**
C_* level	C_tentacle	C_EC_foot		
Value	**1**	**1**		
L_* level	L_foot			
Value	**1**			

At this point the gene is not "unique" in the ordinary sense. It has four leaf assignments and two C_* assignments. Whether those assignments survive depends on which script and threshold is used.

3. What survives the uniqueness filter?

This is where the two scripts differ.

Script	Non-unique rule	Effect on Gene X
get_cluster.R	sum within a hierarchy block > 1	4 leaf assignments are removed; 2 C_* assignments are also removed; L_foot survives if it is the only L_* assignment.
analyze_deseq.R	sum within a hierarchy block > 2	4 leaf assignments are removed; 2 C_* assignments survive; L_foot survives.

Result specifically for unique_celltypes.table

Because **analyze_deseq.R** uses **> 2**, the worked example behaves as follows:

Hierarchy block	Count before filter	Condition	After filter
Leaf cell types	**4**	4 > 2 -> remove	En_Tentacle = NA Ec_Tentacle = NA Ec_BasalDisk = NA Ec_Peduncle = NA
C_* superclasses	**2**	2 > 2 is FALSE	C_tentacle = 1 C_EC_foot = 1
L_* categories	**1**	1 > 2 is FALSE	L_foot = 1

Bottom line: yes - a gene can lose all of its specific leaf-cell-type annotations but still remain annotated to broader categories such as C_tentacle, C_EC_foot, and L_foot. In analyze_deseq.R, two simultaneous C_* assignments are retained because the removal threshold is more than two, not more than one.

Why this matters

The filename **unique_celltypes.table** can be misleading. Under the current analyze_deseq.R logic, a gene may still have two annotations within the same hierarchy block. The table is therefore better understood as a hierarchy-filtered annotation table rather than a strictly one-cell-type-per-gene table.

Source code used for this explanation

File	Relevant role
scripts/dropseq_final_atlas_analysis.R	Derives gene x cell-type mean expression from Seurat atlas identities
scripts/create_sc_atlas_cluser_table.ipynb	Assigns genes to leaf cell types using the above-mean criterion
scripts/get_cluster.R	Creates C_* and L_* categories and uses >1 for its own unique table
scripts/analyze_deseq.R	Creates unique_celltypes.table and uses >2